

Principles of Economics

Week 2: Scarcity, Work and Technology

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Plan for today's class

Work or leisure?

The work-leisure choice model

The feasible set

Preferences

The optimal choice

Comparative statics

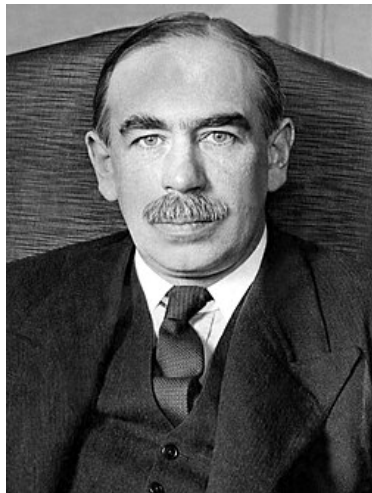
Work or leisure?

Summary

1. Work or leisure?

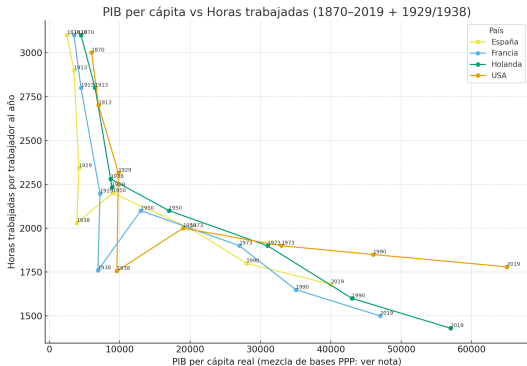
Keynes (1930): *Economic Possibilities for our Grandchildren*

- ▶ Predicted a 15-hour work week
- ▶ Based on productivity growth
- ▶ More free time for leisure and culture
- ▶ Is this true today?
- ▶ Cutting the working day to 7 hours?
- ▶ Or a 4-day week?



Has the rise in living standards affected work?

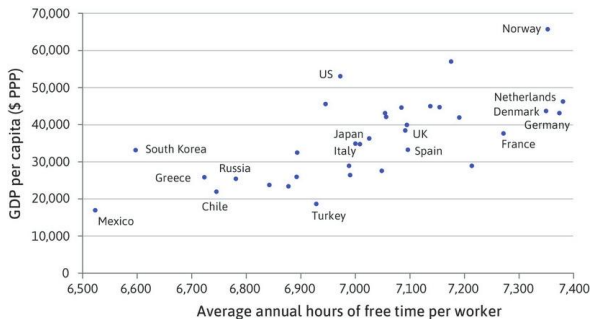
- ▶ Many countries: **explosive growth since 1870.**
- ▶ Richer countries: did people work more?
- ▶ **Leisure time:** also part of living standards.



- ▶ As a benchmark:
40h/week*52 weeks = 2080 hours/year.
- ▶ In the Netherlands (1870-2000):
 - ▶ GDP:
$$\frac{22k - 2.8k}{2.8k} = +704\%$$
 - ▶ Hours worked:
$$\frac{1.4k - 3.3k}{3.3k} = -58\%$$

Has the rise in living standards affected work?

- ▶ Positive correlation: GDP per capita and leisure time.
- ▶ Similar GDP, very different work-leisure balance (South Korea vs France).



Today: an individual choice model for differences in work and leisure hours, across countries and over time.

Source: Annual hours of free time per worker and income (2013). core-econ (Micro, Unit 3)

2. The work-leisure choice model

A model of work and leisure time

Elements

Two fundamental elements:

What are individuals able to do?

How much effort can they put into their work?

What is the outcome of that effort?

The role of productivity



Tools:

Production function and feasible set



Constraints



What are individuals willing to do?

What provides utility/satisfaction to the consumer?

How are the different combinations valued?

The role of scarcity.



Tools:

Individual preferences, indifference maps



What they want



Choice:

Work and leisure

3. The feasible set

A model of Work and Leisure Time

What can be done: the production function

- ▶ Limits to choice: not all options are possible.
 - ▶ No more than 24 hours a day (less, counting sleep).
- ▶ A model of a student's decision: study hours and the exam grade.
 - ▶ A tool relating study effort to the grade?
 - ▶ Yes! The production function, $Y=f(X)$, where:
 - ▶ Y is the exam grade (output);
 - ▶ X is the study time (input);
 - ▶ The student is the technology that transforms effort into a grade

Review the previous slides for the technical assumptions.

A model of Work and Leisure Time

What can be done: an example – study time and grades

- ▶ Studies show a positive relationship between study time and grades:

Average grade	A lot of study time	Little study time
Suitable environment	3,6	3,4
Unsuitable environment	3,4	3,2

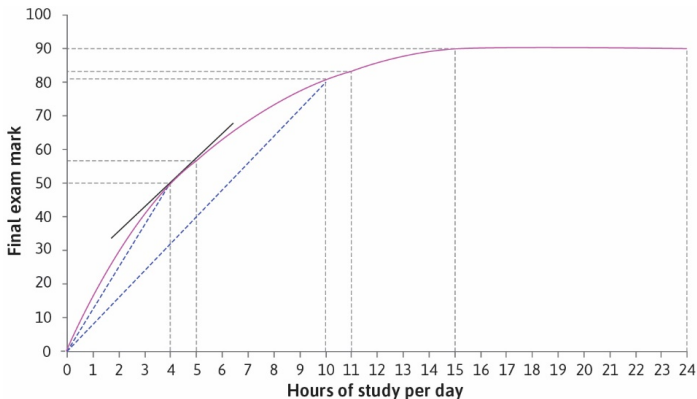
Source: [Plant et al](#) (Contemporary Educational Psychology, 2005).

- ▶ Assumption: more study, higher grade (positive relationship).
- ▶ And diminishing marginal returns:
 - ▶ Studying little: one more hour, big gain in the grade;
 - ▶ Studying a lot: still positive, but smaller.

A model of Work and Leisure Time

What can be done: an example – study time and grades

- Production function relating study time to the exam grade:



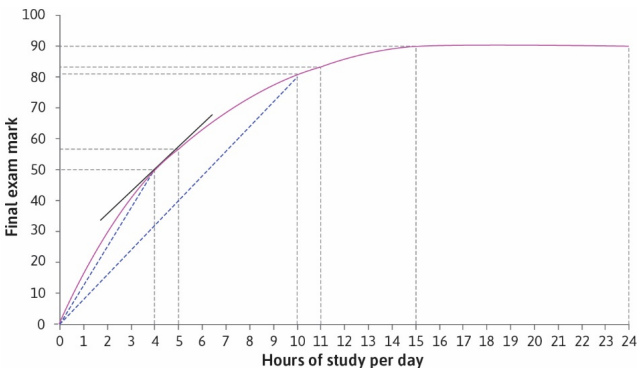
Study hours (input)	Grade (output)
0	0
1	20
2	33
3	42
4	50
5	57
6	63
7	69
8	73
9	78
10	81
11	84
12	86
13	88
14	89
≥ 15	90

- Again, everything else constant (*ceteris paribus*)

A model of Work and Leisure Time

What can be done: average and marginal product

- Production function relating study time to the exam grade:



Two key concepts:

- **Marginal product:** change in output from changing the input by one unit (at a given point, holding the other factors constant);
- **Average product:** average output per unit of input.

A model of Work and Leisure Time

What can be done: average and marginal product

- Marginal and average product from the table:

- **Marginal product:**

$$\frac{\Delta y}{\Delta h} = \frac{y_2 - y_1}{h_2 - h_1} = \frac{f(h_2) - f(h_1)}{h_2 - h_1}$$

$(h_1, f(h_1))$ is the initial point of the production function.

- **Average product:**

$$\frac{y}{h} = \frac{f(h)}{h}$$

$y = f(h)$: Grade [output]

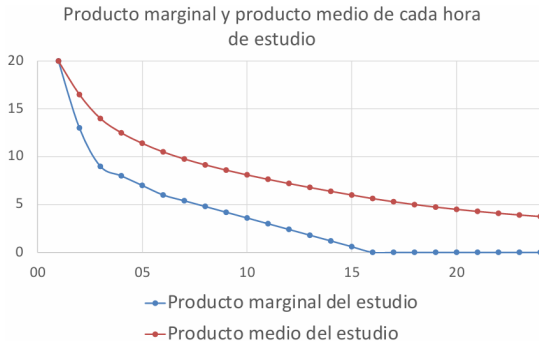
h : study hours [input]

Study hours (input)	Grade (output)	Average product	Marginal product
0	0		
1	20	$\frac{20}{1} = 20$	$\frac{20-0}{1-0} = 20$
2	33	$\frac{33}{2} = 16.5$	$\frac{33-20}{2-1} = 13$
3	42	$\frac{42}{3} = 14$	$\frac{42-33}{3-2} = 9$
4	50	12.5	8
5	57	11.4	7
6	63	10.5	6
7	69	9.9	6
8	73	9.1	4
9	78	8.7	5
10	81	8.1	3
11	84	7.6	3
12	86	7.2	2
13	88	6.8	2
14	89	6.4	1
15	90	6	1

A model of Work and Leisure Time

What can be done: average and marginal product

- Average and marginal product, plotted:



- **The marginal product is decreasing:** the more you study, the less productive each hour.
- The marginal product curve lies below the average product curve. Why?

What can be done: opportunity cost and the feasible frontier

- ▶ **Opportunity cost:** (other textbooks define it differently).

The opportunity cost of an action is the net benefit of the next best alternative action.

- ▶ Opportunity costs matter when comparing **mutually exclusive alternatives**:
 - ▶ Reflected in the economic cost.
 - ▶ **Economic costs** can be defined as:
 - ▶ Economic cost = monetary cost + **subjective costs** (e.g. transport cost + effort of work + opportunity cost of forgoing the best alternative);
- ▶ Benefit above economic cost: **economic rent**.

Economic rent is usually different from (accounting) profit!

A model of Work and Leisure Time

What can be done: opportunity cost and the feasible frontier

- **Example:** you choose between A) a play at the theatre and B) a concert.

		Low value of the concert	High value of the concert
(1)	Price of the theatre ticket. (monetary cost)	25	25
(2)	The pleasure lost by not going to the concert. (subjective/opportunity cost)	15	30
(1) + (2)	Economic cost	40	55
(3)	Benefit of going to the theatre (economic benefit)	50	50
(3) - (2)	Economic rent	10	-5

- Missing the concert costs us little: we choose A.
- Higher opportunity cost (*ceteris paribus*): we may choose B.

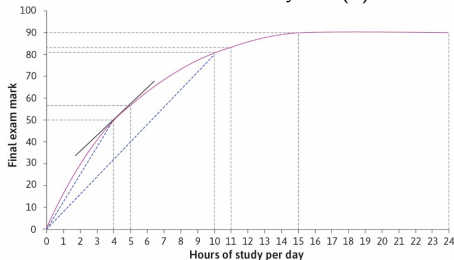
The benefit (and monetary cost) of the theatre does not change; what we give up does.

A model of Work and Leisure Time

What can be done: opportunity cost and the feasible frontier

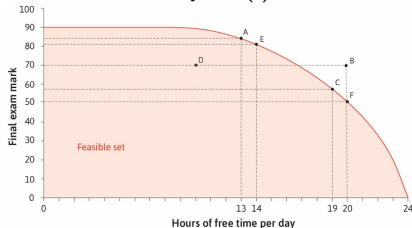
- The **feasible frontier**: **maximum output** with a given amount of input:

Production function $y = f(h)$



Daily study hours

Feasible frontier $y = f(l)$, l = leisure



Daily hours of free time

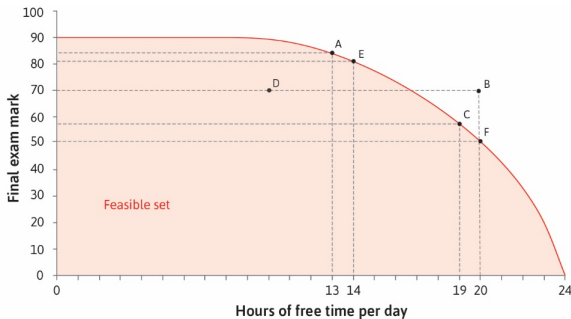
- Study 10 hours $\rightarrow$ 14 of free time ($10+14=24$) $\rightarrow$ grade 80% (point E)

Horizontal axis: now hours of free time ($=24\text{h}-\text{hours worked}$). The feasible frontier is a mirror image of the production function.

A model of Work and Leisure Time

What can be done: opportunity cost and the feasible frontier

- ▶ The **feasible frontier**: maximum output with a given amount of input:



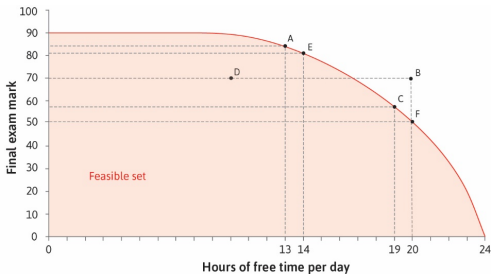
- ▶ B is **unattainable** (technologically impossible)
- ▶ D is **wasteful** (with that amount of free time we could produce more)
- ▶ A, E, C, F are on the **efficient frontier** (no waste)

Horizontal axis: hours of free time (=24-hours worked).

A model of Work and Leisure Time

What can be done: opportunity cost and the feasible frontier

- The **feasible frontier**: maximum output with a given amount of input:



The Marginal Rate of Transformation (MRT) is the slope of the feasible frontier: the trade-off an individual faces [here, the opportunity cost of free time].

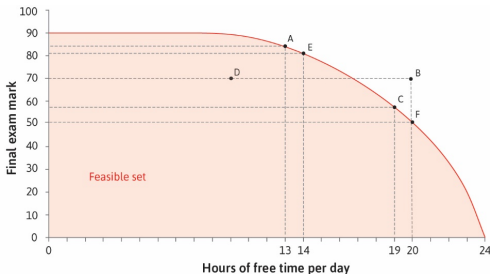


On the frontier you face a trade-off: more of one good (grades) means less of another (free time): opportunity cost.

A model of Work and Leisure Time

What can be done: opportunity cost and the feasible frontier

- The **feasible frontier**: maximum output with a given amount of input:



The **Marginal Rate of Transformation (MRT)** is the slope of the feasible frontier: the trade-off an individual faces [here, the opportunity cost of free time].

- **From A to E**: 1 extra hour of free time, 3 grade points lost (opportunity cost):
 $MRT_A = 3$
- **From C to F**: Even greater opportunity cost.
 $MRT_C = 7$

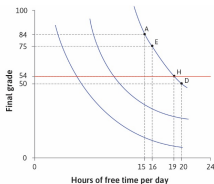
The MRT increases with free time (diminishing marginal returns).

4. Preferences

A model of Work and Leisure Time

What individuals are willing to do: preferences and decisions under scarcity

- ▶ The student gets utility from grades and from free time. **Their preferences:** an indifference map.



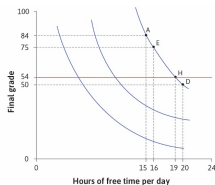
- ▶ **Points A, E, H and D** give the same satisfaction (same IC, no ranking):
 - ▶ The student is indifferent:
 $A \sim E \sim H \sim D$;
- ▶ **Points B and C** give less satisfaction than A (or E, H, D):
 - ▶ They lie on an IC closer to the origin.
 - ▶ We say A is preferable to B: $A \succ B$ (and $A \succ C$).

Indifference curve (IC): all combinations of goods with the same utility [blue lines]

A model of Work and Leisure Time

What individuals are willing to do: preferences and decisions under scarcity

- The student gets utility from grades and from free time. **Their preferences:** an indifference map.



	A	E	F	G	H	D
Hours of free time	15	16	17	18	19	20
Final grade	84	75	67	60	54	50
Marginal rate of substitution between grades and free time (MRS)	9	8	7	6	4	

$9 = 84 - 75$: 9 grade points given up for 1 more hour of free time (15 $\rightarrow$ 16).

The Marginal Rate of Substitution (MRS) is the slope of the indifference curve: the trade-off at constant satisfaction.

What individuals are willing to do: assumptions and properties of preferences

The shape of the indifference map requires assumptions about preferences:

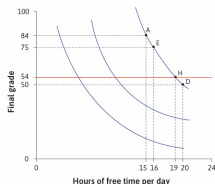
- ▶ **Assumption 1:** more of both goods is better; preferences are transitive (if $A \succeq B$ and $B \succeq C$, then $A \succeq C$).
 - ▶ Implication 1: indifference curves have a negative slope;
 - ▶ Implication 2: northeast (up-right) beats southwest (down-left)
 - ▶ Implication 3: indifference curves do not cross.
- ▶ **Assumption 2:** Indifference curves are smooth. [see the concept of the "spherical cow", about modelling reality.]
- ▶ **Assumption 3:** Indifference curves flatten towards the southeast (down-right).
 - ▶ Implication 4: the MRS decreases (the trade-off between goods depends on relative scarcity).

Note: $\succeq$ means weakly preferred (similar to $\geq$)

A model of Work and Leisure Time

What individuals are willing to do: assumptions and properties of preferences

Assumption 3 and the behaviour of the MRS.



	A	E	F	G	H	D
Hours of free time	15	16	17	18	19	20
Final grade	84	75	67	60	54	50
Marginal rate of substitution between grades and free time (MRS)	9	8	7	6	4	

- ▶ **From A to E:** 9 grade points for one more hour of free time.
 - ▶ You have a high grade/little free time at A
- ▶ **From H to D:** only 4 grade points for one more hour
 - ▶ You have a lot of free time/a low grade at H

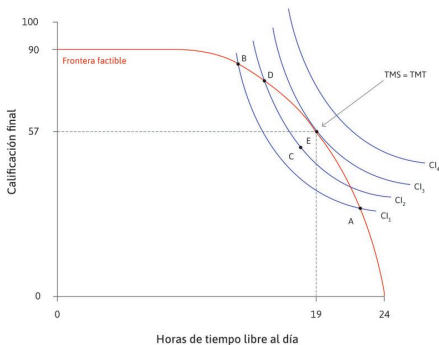
The MRS decreases as free time increases.

5. The optimal choice

A model of Work and Leisure Time

Now putting it all together: the equilibrium model

Constraint (FF) + preferences (IC) $\rightarrow$ equilibrium choice between work and leisure:



- ▶ The solution is the best feasible combination:
 - ▶ Why isn't C optimal?
 - ▶ What about D?
- ▶ So, in equilibrium:
 - ▶ The IC is tangent to the feasible frontier.

$$\text{MRT} = \text{MRS}$$

A model of Work and Leisure Time

Now putting it all together: equilibrium

- ▶ Why does equilibrium require $MRS=MRT$? Remember:
 - ▶ **MRS** = grade I am willing to give up for one less hour of free time;
 - ▶ **MRT** = grade I gain from one less hour of free time
- ▶ So:
 - ▶ $MRS < MRT$ (point A): work more; I give up fewer grade points than I gain in output.
 - ▶ $MRS > MRT$ (point D): work less; I would give up more grade points than I lose in output.
- ▶ Only $MRS=MRT$ is satisfying.

A model of Work and Leisure Time

Now putting it all together: equilibrium

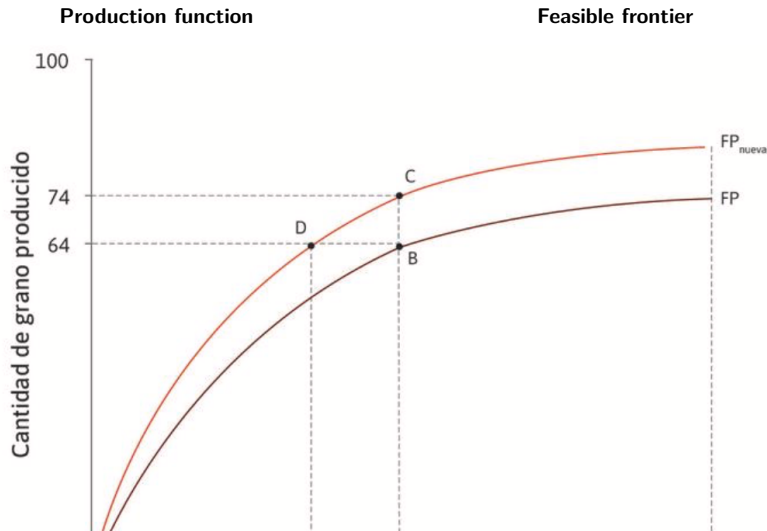
Study effort, free time and grades: a classic **constrained choice** problem:

- ▶ **Student's objective:** maximise satisfaction (depends on free time and grade – scarce goods with an opportunity cost)
 - ▶ They prefer as much of both as possible.
- ▶ **Student's constraint:** the extent to which study effort can be converted into grades.
 - ▶ Technology makes some combinations unattainable.
- ▶ Equilibrium reconciles the two trade-offs:
 - ▶ MRT: trade-off between different combinations in terms of feasible alternatives;
 - ▶ MRS: trade-off between different combinations in terms of satisfaction.

6. Comparative statics

A farmer deciding how much to work to produce food

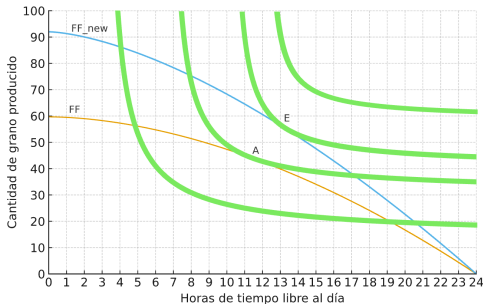
- ▶ **Effect of shocks:** a farmer decides how much to work to produce grain (food).
- ▶ **Production function and feasible frontier** under two technologies:



Comparative Statics

A farmer deciding how much to work to produce food

The farmer's preferences over food and free time:

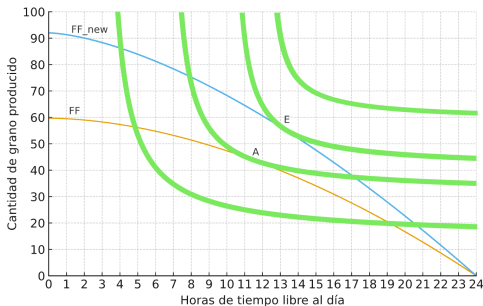


- ▶ **Equilibrium choice** of free time and food, as before: $MRS=MRT$
- ▶ **Initial equilibrium:**
 - ▶ Point A: 16 units of free time and 55 of grain.
- ▶ **The technological improvement expands the feasible set: new equilibrium.**
 - ▶ Point E: 17 units of free time and 61 of grain.
 - ▶ Preferences do not change! (ceteris paribus)
 - ▶ E gives more satisfaction than A – why?

Comparative Statics

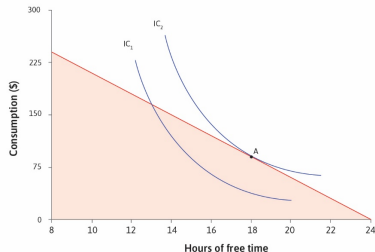
A farmer deciding how much to work to produce food

The farmer's preferences over food and free time:



- ▶ In this case, **when technology improves:**
 - ▶ **Both** free time and the amount of food increase.
 - ▶ Therefore, hours worked decrease.
- ▶ Does an expansion of the feasible set always increase both goods?
 - ▶ *A slight change to the problem makes it easier to solve.*

The worker-consumer's trade-off: free time versus consumption.



Hours worked	0	2	4	6	8	10
Free time, t	24	22	20	18	16	14
Consumption, c (\$)	0	30	60	90	120	150

► Budget constraint:

$$c = w(24 - t) = 15(24 - t) = 360 - 15t$$

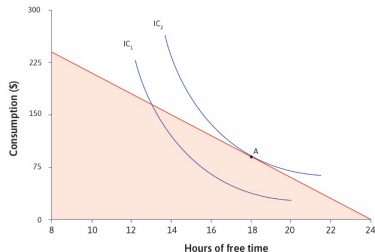
► Budget constraints are the **feasible frontiers for consumption choices**.

► Working more:

- More income, more consumption;
- But less leisure;
- Again, there is a trade-off.

- The **wage** is constant: $w = 15$.
- Max **hours worked**: 16.
- Min **leisure time**: $t = 8$.

The worker-consumer's trade-off: free time versus consumption.



Hours worked	0	2	4	6	8	10
Free time, t	24	22	20	18	16	14
Consumption, c (\$)	0	30	60	90	120	150

► Again, in equilibrium:

$$MRS = MRT = wage$$

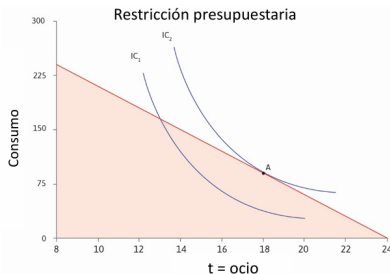
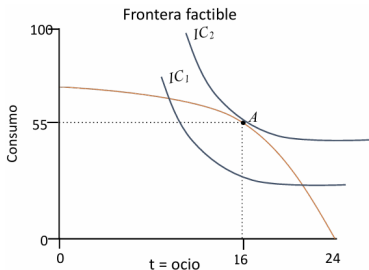
The MRT is the slope of the feasible frontier, now the budget constraint, with slope w .

► This happens at **point A**, so the worker-consumer:

- The **wage** is constant: $w = 15$.
- Max **hours worked**: 16.
- Min **leisure time**: $t = 8$.
- Spends \$90 on consumption and
- Has 18 hours of free time (works 6 hours)

Feasible frontier versus budget constraint

Perspective



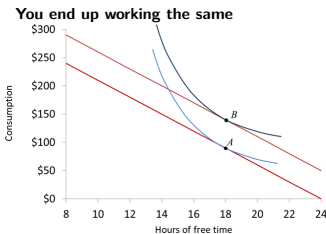
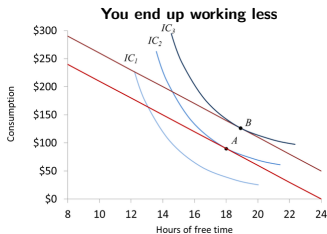
- ▶ Mirror image of the **production function** $Y = f(t)$
- ▶ **Concave** due to diminishing marginal returns.
- ▶ More work, smaller increases in output.

- ▶ Given by **wages**:
 $c = wh = w(24 - t) = -wt + 24w$
- ▶ **Linear** because the MRT is the wage.
- ▶ Work increases consumption at a constant rate.

Changes in in the budget

Total effect of an increase in the wage

- ▶ **The feasible frontier shifts out in parallel** (say, a technology shock): same free time, more output and consumption. **Effect on the equilibrium?**

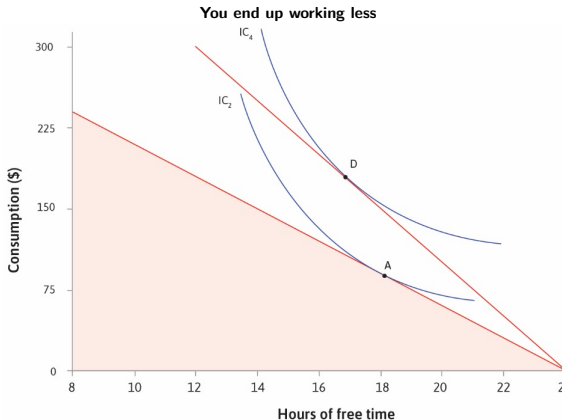


A: original equilibrium. B: final equilibrium, after the expansion.

- ▶ **Equilibrium consumption increases.**
- ▶ **Equilibrium free time can:**
 - ▶ Increase
 - ▶ Not change
 - ▶ Or even decrease. But why?

Not every shock shifts the FF in parallel. **Wage increase:**

- ▶ What happens **to the feasible frontier**?
 - ▶ Same hours, more consumption.
 - ▶ It expands too, but rotating upward.
 - ▶ Remember the FF is given by $c = w(24 - t)$
- ▶ Graphically:
 - ▶ **Initial equilibrium: A**
 - ▶ The wage increase causes an **upward rotation of the FF**;
 - ▶ **Final equilibrium: D**
 - ▶ New equilibrium: more consumption, less rest.



What explains **the total effect of a wage increase ($\uparrow w$) on the hours worked/free time relationship?**

- ▶ Two separate effects. A wage increase creates two incentives:
 - ▶ Work less [more free time]: you are paid more per hour (Income Effect – IE)
 - ▶ Work more [less free time]: the opportunity cost of not working rises (Substitution Effect – SE)
- ▶ Both act together and in opposite directions, so in general:

Shock	Dominant effect	Overall incentive on work	Total effect on free time
Wage increase ($\uparrow w$)	$ SE < IE $	less	Free time <i>increases</i> ($\uparrow t$)
	$ SE = IE $	same	Free time <i>does not change</i>
	$ SE > IE $	more	Free time <i>decreases</i> ($\downarrow t$)

- ▶ The outcome depends on preferences.

Same frontier, different choice



Taylor Swift

The Eras Tour, 2023-24



Adele

Las Vegas 2022-24 + Munich 2024

	Swift	Adele
Concerts	149	110
Cities	51, on 5 continents	2
Tickets sold	10.2 m	1.1 m

Same feasible frontier and different indifference curves?

Gross: Eras \$2,078 m; Adele in Las Vegas \$220 m (est.).

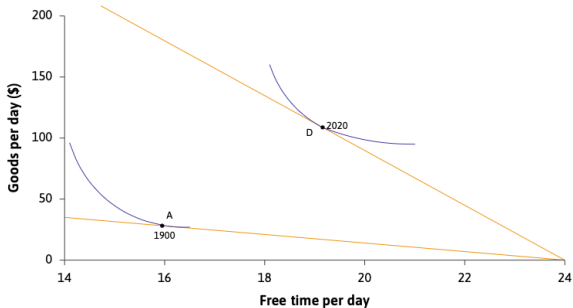
Photos: Paolo V (CC BY 2.0), Oroborvs (CC BY-SA 4.0), via Wikimedia Commons.

7. Work or leisure?

Back to where we started

Explaining changes in hours worked over time

United States: rising wages, less working time.

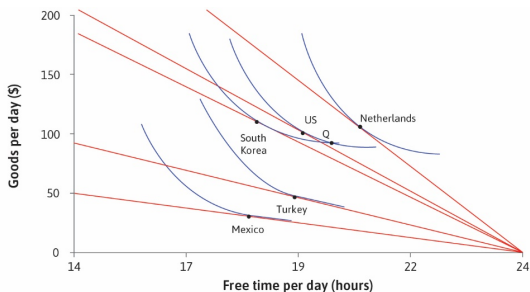


- ▶ In 2020, compared with 1900,
 - ▶ More consumption (higher living standards)
 - ▶ Hours worked have decreased.
- ▶ Why?
 - ▶ The income effect dominated: stronger incentive to work less.

Back to where we started

Explaining changes in hours worked over time

Differences between countries stem from different causes:



- ▶ Different countries have **different**:
 - ▶ Feasible frontiers (differences in technology, education, etc.);
 - ▶ Preferences over free time and consumption.
- ▶ Let's compare the Netherlands and:
 - ▶ Mexico: **different budget set**;
 - ▶ The US: **a weaker taste for free time**.
- ▶ Also culture (norms) and policy (legal limits).

Is this a good model?

Discussion

- ▶ On the one hand:
 - ▶ In reality, people don't make calculations in terms of MRS/MRT;
 - ▶ Most people cannot choose their working hours;
- ▶ On the other hand:
 - ▶ Over time, people learn which combination suits them best;
 - ▶ People choose which jobs to apply for;
 - ▶ The model helps us understand real-world phenomena:
 - ▶ Preferences and income/substitution effects explain individual choices
 - ▶ And differences in working hours across countries and over time.

8. Summary

1. **Work or leisure?** → richer countries, fewer hours worked: leisure is part of living standards.
2. **The work-leisure choice model** → what individuals can do (constraints) and what they are willing to do (preferences).
3. **The feasible set** → production function, diminishing returns, opportunity cost; the MRT is the slope of the frontier.
4. **Preferences** → indifference maps and their assumptions; the MRS falls as free time rises.
5. **The optimal choice** → constrained optimisation: the IC is tangent to the frontier, $MRS=MRT$.
6. **Comparative statics** → technology and wage shocks; income versus substitution effect explains hours worked.

Key concepts: production function, average and marginal product, opportunity cost, economic cost and rent, feasible frontier, MRT, indifference curve, MRS, budget constraint, income and substitution effects.